LEF MODEL · CHAPTER 4

Chapter 4: The Skin That Rises from Outside — the 4 Stages of the Dome and the Translation Rule of the Moon's Face

Based on Paper 03 — LEF Mechanism in Practice (Part B: §B.1–B.7)

01 · THE MYSTERY

The mystery: where did the thick far-side crust come from — and who carved the ancient craters into it?

In the previous chapter we discovered how the Earth's internal engine pushes deep, foamed magma upward at a pressure of 150.26 GPa. But even perfect internal feeding leaves three great riddles on the surface of the Moon that we see in telescopes and in the gravity maps of the GRAIL spacecraft:

1. **Why is the crust on the far side much thicker than on the near side?** On the far side the crust reaches an average thickness of about 55 km, while on the near side it is only about 30–35 km thick, and under the great maria it is thinner still.
2. **Why are the Moon's oldest craters spread over both hemispheres, and even especially crowded around the poles?** Had the Moon coalesced in space from a scorching debris disc, its first surface should have been a smooth, craterless magma ocean.
3. **Why do the two hemispheres look so different from each other** — the near side covered with dark, round basaltic maria, while the far side is a bright, crater-scarred highland?

The LEF model's answer is surprising in its simplicity: **the Moon was not built only from within — it was built at the same time from outside as well**. The light outer crust of the young Earth (the **sial** layer) that lay above the building zone was neither torn nor left behind; it **rose together with the developing body, slid to its tip and became the Moon's outer skin!**

02 · AN IMAGE

An image: the balloon that inflates under a sheet on the ground — and the ball that shrinks a map

To picture how a flat cover on the Earth's surface turns into a whole sphere with three defined zones, imagine two simple experiments:

- **The balloon that inflates under a fabric sheet (stages 1 and 2):** lay a thick, flexible fabric sheet (representing the Earth's outer sial layer) on the ground, and connect a feeding tube under its centre. When the tube pumps air and material inward, the sheet does not stay flat on the ground — it rises as a dome, and gathers inward from the edges toward the centre, like a tablecloth lifted from its middle. Every mark or engraving already drawn on the sheet while it lay flat on the ground (the ancient craters!) rises with it to the top of the dome.
- **A squashed sponge ball that inflates and rounds outward (stage 3 — the translation rule):** take a half-empty sponge ball and press it from above onto the round head of a round stand (the "tip of the pear"). The lower part of the ball wraps the stand like a wide cap, and the upper part rises as a dome. Now inflate the ball gradually so that it becomes a perfect round sphere: as it rounds outward, **its lower edges peel off the stand from the outside inward**, until only a small circle at the centre of the bottom remains stuck to the stand — and every part of the ball's surface keeps exactly its relative share of the surface area!

Science and model: the four stages of the outer building — from the plain of the Pacific to the foamed dome

Part B of Paper 03 follows, step by step, the Earth's outer crustal layer — the **sial** layer (Sial: a light rock rich in silicon and aluminium, at a density of about 2,800 kg/m³), lying on top of the denser **sima** layer (Sima: rich in silicon and magnesium) — through four continuous stages:

Stage 1 — the flat stage and the gathering of the crust (about 4.5 billion years ago)

Before the body rises, the sial layer lies flat on the Earth's surface above an active feeding zone. Through openings in the crust, deep material rises and is ejected; part of the ejected material falls back onto the exposed crust and creates craters in it (to be explored in depth in the crater chapters). The meaning is astonishing: **the Moon's oldest surface was not formed on the Moon — it was formed on the Earth's surface even before the Moon separated from it!**

In the geographic reconstruction presented in the paper, the crustal material gathered inward into the dome came from the region of the ancient Pacific basin (an area of about 1.0–1.65 × 10^{14} m², 1.4 to 2.4 times the area of the upper dome in the next stage). The base of the pear stood in the east, near the island of Kiritimati in Kiribati (1.9°N, 157.5°W), above the slow province at the bottom of the mantle; and because of the Earth's rotation and oblique feeding from north of the equator, the pear leaned backward (westward) and northward toward latitude 22.5°N.

Stage 2 — the squashed body on the tip of the pear (the foamed state)

When the full mass of the Moon (7.35 × 10^{22} kg) has accumulated at the top of the pear, it does not float as a free sphere but sits as a squashed body made of two domes on a wide pear tip **6,000 km** across:

- **The lower dome (33% of the surface area):** hugs from above the convex tip of the pear (whose radius of curvature is about 4,000 km and which reaches

about 1,350 km into the base of the body).

- **The upper dome (67% of the surface area):** rises to a height of about **3,650 km** above the contact edge and is exposed to space.

The perfect link between shape and pressure (density 1,300 kg/m³)

With a dome height of 3,650 km over a base of 6,000 km, the total volume of the body is 5.65 × 10^{19} m³ (2.5 times the Moon's volume today). Dividing the Moon's mass by this volume gives a density of **1,300 kg/m³** — exactly the density of foamed, gas-bubble-rich magma that the pressure test of Chapter 3 required in order to lift the material! The high dome not only fits the internal engine, it is also carried upward on a "lubricated", weak shear layer between the sial and the sima, while a cold, stiff outer shell ($\eta \gtrsim 10^{23}\ \mathrm{Pa \cdot s}$) protects the craters on it from collapse.

04 · THE MECHANISM IN WORDS

The mechanism in words: the translation rule, the exposure of the maria and the building of the crust in layers

When the self-gravity of the rising body overcomes the local tidal stretching, the body passes into **stage 3 — the outward rounding**: the squashed body inflates and rounds into a sphere at constant volume (the surface area shrinks by a factor of 0.68, just as a sphere holds the minimum surface area for a given volume). The contact with the pear narrows from a width of 6,000 km to a connecting disc of **3,000 km**, and all the rest of the lower dome gradually peels off from the outside inward, together with the sial layer on which it sat.

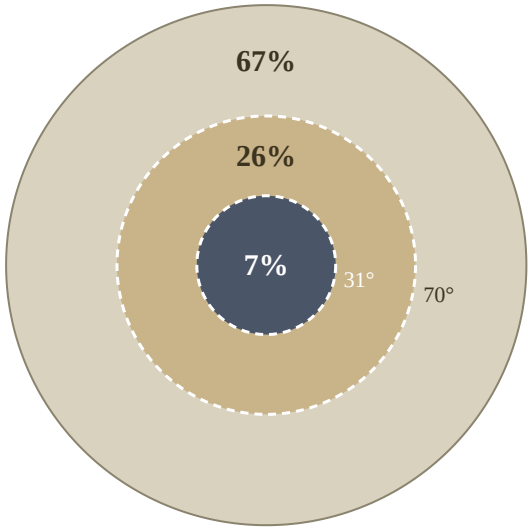
From this follows the **geometric translation rule**: in the outward rounding, **every part keeps its relative share of the surface area**. So the Moon's face divides into three concentric zones around the centre of contact (24.1°N, 335.2°E):

1. **The near-side cap (0°–31°, about 7% of the area — about 1,900 km across):** this is the last contact disc (3,000 km) that stayed attached to the pear until

the moment of release. Through it passed the central internal feeding conduits — **Mare Serenitatis** (which peeled off first) and **Mare Imbrium** (which stayed attached until the moment of final detachment!).

2. **The peeling belt (31°–70°, about 26% of the area):** this is the ring of the lower dome that gradually peeled off during the outward rounding and carried with it the sial layer on which it sat.

3. **The far side (70°–180°, about 67% of the area):** this is the upper dome that was exposed to space from the first moment, and carried its ancient craters up with it.



The near-side cap · 0°–31° · 7%
The last contact disc; Serenitatis and Imbrium

The peeling belt · 31°–70° · 26%
The ring of the lower dome that peeled off with the sial

The far side · 70°–180° · 67%
The upper dome, exposed to space from the start

The three zones around the centre of contact (24.1°N, 335.2°E) on an equal-area map: the area of each ring on the map equals its share of the Moon's surface.

In **stage 4** (flattening by the rotation and inward rounding while the body is compacted), the stiff crust keeps the size of the existing craters while the surfaces between them draw together — which crowds the craters around the poles. And at the same time, throughout the whole journey, the firing of material from the active openings and its return to the upper dome continued: this is how the thick far-side crust was built **layer upon layer**, from sial that migrated to the tip and from material that came back and landed on it again and again!

05 · THE KEY EQUATIONS

Academic explanation and the key equation

The passage from a squashed dome to a sphere with three concentric bands is defined by the conservation of surface-area ratios ($f_i = A_i/A_{\text{tot}}$). For a spherical cap with a central angle θ from the centre of contact, its relative share of the total sphere's area ($4\pi R^2$) is:

$$f(\theta) = \frac{2\pi R^2(1-\cos\theta)}{4\pi R^2} = \frac{1-\cos\theta}{2} \quad \Longrightarrow \quad \theta = \arccos(1-2f)$$

Inserting the area shares of stage 2 ($f_1 = 0.07$ for the 3,000-km contact disc, and $f_1 + f_2 = 0.33$ for the whole 6,000-km lower dome) gives at once the exact angles on the Moon: $\theta_1 \simeq 31^\circ$ and $\theta_2 \simeq 70^\circ$. At the same time, the mass balance of the crust closes completely: the far-side crust (average thickness 55 km at a density of 2,550 kg/m³) weighs about 2.6 × 10²¹ kg (3.5% of the Moon's mass), and the near-side crust (30–35 km) weighs about 1.5 × 10²¹ kg — together about 4%–6% of the Moon's mass, while the remaining 94%–96% is deep mantle material supplied from within.

06 · THE SIGNATURE

The signature in reality: the GRAIL crustal-thickness map and the nesting of the maria

When the two circles of the geometric translation rule (31° and 70° around 24.1°N, 335.2°E) are drawn on the Moon's real crustal-thickness map measured by the **GRAIL** spacecraft and on the **LOLA** surface topography, a remarkable match appears:

- **A precise geographic nesting of the maria:** the outer circle (6,000 km at the time of building) contains **99%** of all the basaltic maria of the near side; the inner contact circle (3,000 km) contains **66%** of them; and Mare Imbrium lies right inside the inner contact circle.
- **The reversal of crustal thickness between the sides:** at the centre of the near side, beneath the maria that served as feeding openings and lost their

sial layer to the heat of contact, the crust is especially thin (blue in the GRAIL maps); while beyond the 70° boundary, on the far side, rise the ancient, thick highlands built in layers.

- **A prediction of the composition with depth:** since internal sorting in the Earth progressed with time, the earliest layers laid down at the bottom of the far-side crust were less sorted and richer in iron, while the upper, later layers are light and poor in iron — a prediction that can be tested directly in the central peaks of impact craters, which expose rock from different depths!

The lab box: the 4 stages of the dome, the translation rule and the crustal balance (Paper 03, Part B)

The zone at the time of building (stage 2)	Surface area	Relative share	Angle from the centre of contact on the Moon	The geographic identity on the Moon today
The contact disc (3,000 km)	**7.3 × 10^{12} m^2**	**7%**	**0° – 31°**	**The near-side cap (~1,900 km across); contains 66% of the maria (Serenitatis and Imbrium)**
The peeling ring (up to 6,000 km)	2.7 × 10^{13} m^2	**26%**	**31° – 70°**	The transition belt — the sial that peeled off the pear; together with the cap it contains 99% of the maria
The upper dome (height 3,650 km)	7.0 × 10^{13} m^2	**67%**	**70° – 180°**	The far side — a thick crust (~55 km) built of layers of sial and returning material, ancient in its craters

State of the body (constant lunar mass)	Geometry and dimensions	Total volume	Mean density
On the tip of the pear (stage 2)	Base 6,000 km, dome height 3,650 km	5.65 × 10^{19} m^3	**1,300 kg/m^3** (foamed state)
A sphere at the moment of release (stage 4)	Radius 1,890 km	2.83 × 10^{19} m^3	**2,600 kg/m^3** (rigid shell)
The Moon today	Radius 1,737 km	2.20 × 10^{19} m^3	**3,344 kg/m^3** (compacted)

07 · NEXT

The door ahead

We have seen how the Earth's outer skin rose with the body, gradually peeled off and organised itself into the three bands of the Moon's face — and how the combination of internal feeding and the outer dome created the differences in thickness between the near side and the far side.

But now the most delicate and dramatic engineering question remains: **what bridge connected this dome to the Earth without breaking under the shear stresses of latitude 22.5°, and how did it separate exactly 210 km beyond the corotation line?** In the next chapter — **Chapter 5: The Pear and the Stefan Release — the Bridge That Carries the Moon and Tears Exactly at the Corotation Line** (Paper 03, Part C: §C.1–C.6) — we will examine the separation mechanism itself.

Chapter 5

The Pear and the Stefan Release — the Bridge That Carries the Moon and Tears Exactly at the Corotation Line →

← Chapter 3 · Prologue

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